Nivolumab 3mg/kg with Ipilimumab 1mg/kg Therapy

INDICATIONS FOR USE:

INDICATION	ICD10	Regimen Code	HSE approved reimbursement status*
Nivolumab in combination with ipilimumab for the first-line treatment of adult patients with intermediate/poor-risk advanced renal cell carcinoma (RCC).	C64	00551a	ODMS Ipilimumab: 01/02/2021 Nivolumab: 01/02/2021 (IV) 11/02/2026 (SC)
Nivolumab in combination with ipilimumab is indicated for the treatment of advanced (unresectable or metastatic) melanoma in adults[i].	C43	00551b	ODMS Ipilimumab: 01/10/2020 Nivolumab: 01/10/2020 (IV) 11/02/2026 (SC)
Nivolumab in combination with ipilimumab for the treatment of adult patients with mismatch repair deficient or microsatellite instability-high metastatic colorectal cancer (MSI-H mCRC) after prior fluoropyrimidine-based combination chemotherapy.	C18, C19, C20	00551c	ODMS Ipilimumab: 01/06/2023 Nivolumab: 01/06/2023 (IV) 11/02/2026 (SC)

* This applies to post 2012 indications

TREATMENT:

The starting dose of the drugs detailed below may be adjusted downward by the prescribing clinician, using their independent medical judgement, to consider each patient's individual clinical circumstances.

For RCC and melanoma indications (00551a/00551b):

Nivolumab and ipilimumab are administered on Day 1 of a 21-day cycle for the first 4 cycles.

From cycle 5, nivolumab is administered as monotherapy at a dose of 240mg IV or 600mg SC on Day 1 of a 14-Day cycle (Refer to NCCP Regimen 00483) or at 480mg IV or 1200mg SC on Day 1 of a 28-Day cycle (Refer to NCCP Regimen 00484) until patient experiences disease progression or unacceptable toxicity.

For the monotherapy phase, the first dose of nivolumab should be administered:

- 3 weeks after the last dose of the combination of nivolumab and ipilimumab if using 240mg IV or 600mg SC every 14 days schedule

or

- 6 weeks after the last dose of the combination of nivolumab and ipilimumab if using 480mg IV or 1200mg SC every 28 days schedule

The information contained in this document is a statement of consensus of NCCP and ISMO or IHS professionals regarding their views of currently accepted approaches to treatment. Any clinician seeking to apply or consult these documents is expected to use independent medical judgement in the context of individual clinical circumstances to determine any patient's care or treatment. Use of these documents is the responsibly of the prescribing clinician. and is subject to HSE's terms of use available at http://www.hse.ie/eng/Disclaimer

This information is valid only on the day of printing, for any updates please check www.hse.ie/NCCPSACTRegimens

For dMMR/MSI-H mCRC after prior fluoropyrimidine-based combination chemotherapy indication (00551c):

Nivolumab and ipilimumab are administered on Day 1 of a 21-day cycle for the first 4 cycles.

From cycle 5, nivolumab is administered as monotherapy at a dose of 240mg IV or 600mg SC on Day 1 of a 14-Day cycle (Refer to NCCP Regimen 00483), or 1200mg SC on Day 1 of a 28-Day cycle (Refer to NCCP Regimen 00484) unless patient experiences disease progression or unacceptable toxicity.

For the monotherapy phase, the first dose of nivolumab should be administered:

- 3 weeks after the last dose of the combination of nivolumab and ipilimumab

Switching monotherapy (i.e. cycle 5 onwards) dosing schedule:

Switching from a 2-weekly to a 4-weekly schedule:

- the first 4-weekly dose (480mg for IV or 1200mg for SC) should be administered two weeks after the last 2-weekly dose (240mg for IV or 600mg for SC)

Switching from 4-weekly to 2-weekly schedule:

- the first 2-weekly dose (240mg for IV or 600mg for SC) dose should be administered four weeks after the last 4-weekly dose (480mg for IV or 1200mg for SC)

Patients should be monitored continuously (at least up to 5 months after the last dose) as an adverse reaction with nivolumab in combination with ipilimumab may occur at any time during or after discontinuation of therapy.

Facilities to treat anaphylaxis MUST be present when systemic anti-cancer therapy (SACT) is administered.

Cycles 1-4

Drug	Dose	Route	Diluent & Rate	Cycle
Nivolumab	3mg/kg	IV infusion	Infuse over 30 minutes through a sterile, non-pyrogenic, low protein binding in-line filter with a pore size of 0.2-1.2 micrometre	Every 21 days for 4 cycles
Ipilimumab	1mg/kg	IV infusion Observe post infusion*	NaCl 0.9% to a concentration between 1 and 4mg/mL over 30 minutes using a 0.2-1.2 micrometre low-protein binding in-line filter	Every 21 days for 4 cycles
Nivolumab or ipilimumab **must not** be administered as an intravenous push or bolus injection.				
Nivolumab can be infused directly as a 10 mg/mL solution or can be diluted to as low as 1 mg/mL with NaCl 0.9% or glucose 5%.				
*Vital signs including temperature, pulse and BP should be taken every 30 minutes for the duration of the infusion and 1 hour following completion of the infusion.				
The line should be flushed with NaCl 0.9% after the ipilimumab infusion has finished.				

Note: Administration volumes and fluids have been standardised to facilitate electronic prescribing system builds.

The information contained in this document is a statement of consensus of NCCP and ISMO or IHS professionals regarding their views of currently accepted approaches to treatment. Any clinician seeking to apply or consult these documents is expected to use independent medical judgement in the context of individual clinical circumstances to determine any patient's care or treatment. Use of these documents is the responsibly of the prescribing clinician. and is subject to HSE's terms of use available at http://www.hse.ie/eng/Disclaimer

This information is valid only on the day of printing, for any updates please check www.hse.ie/NCCPSACTRegimens

Cycle 5 onwards

There are two different formulations of nivolumab available depending on the route of administration.

See **Table 1** and **Table 2** below which refer to intravenous and subcutaneous administration routes respectively.

Table 1: Treatment schedule for Nivolumab (IV) Monotherapy

Drug	Dose	Route	Diluent & Rate	Cycle
Nivolumab	240mg	IV infusion*	Infuse over 30 minutes through a sterile, non-pyrogenic, low protein binding in-line filter with a pore size of 0.2-1.2 micrometres	Every 14 days ongoing to progression or toxicity
OR				
Nivolumab	480mg[a]	IV infusion*	Infuse over 60 minutes through a sterile, non-pyrogenic, low protein binding in-line filter with a pore size of 0.2-1.2 micrometres	Every 28 days[a] ongoing to progression or toxicity
*See alternative treatment schedule for nivolumab SC below.				
[a]To note, the nivolumab 28-day monotherapy IV schedule is currently licensed for the RCC and melanoma indications only.				

ALTERNATIVE TREATMENT SCHEDULE:

Table 2: Treatment schedule for Nivolumab (SC) Monotherapy

Drug	Dose	Route	Diluent & Rate	Cycle
Nivolumab	600mg[a]	SC [c] *	Over 3-5 minutes	Every 14 days ongoing to progression or toxicity
OR				
Nivolumab	1200mg[b]	SC [c] *	Over 3-5 minutes	Every 28 days ongoing to progression or toxicity
[a] Allow one 600mg vial to reach room temperature, then withdraw 5 mL of solution for injection into the syringe.				
[b] Allow two 600mg vials to reach room temperature, then withdraw 10 mL of solution for injection into the syringe.				
[c] May be administered subcutaneously using a 23G-25G hypodermic injection needle or subcutaneous administration set (e.g., winged/butterfly) The hypodermic injection needle must be attached to the syringe immediately prior to administration to avoid clogging. The dose should not be split between two syringes or between two sites of administration. Inject into the subcutaneous tissue of the left or right thigh or abdomen. Alternate injection sites for successive injections. Do not inject at other sites of the body. Do not inject into areas where the skin is tender, red, or bruised, or areas where there are scars or moles. If the administration of nivolumab SC injection is interrupted, it can be resumed at the same site, or at an alternate site. During the treatment course with nivolumab SC, other medicinal products for subcutaneous administration should preferably be injected at different sites.				
*See alternative treatment schedule for nivolumab IV above.				

The information contained in this document is a statement of consensus of NCCP and ISMO or IHS professionals regarding their views of currently accepted approaches to treatment. Any clinician seeking to apply or consult these documents is expected to use independent medical judgement in the context of individual clinical circumstances to determine any patient's care or treatment. Use of these documents is the responsibly of the prescribing clinician. and is subject to HSE's terms of use available at http://www.hse.ie/eng/Disclaimer

This information is valid only on the day of printing, for any updates please check www.hse.ie/NCCPSACTRegimens

ELIGIBILITY:

- Indications as above
- Adequate haematological, hepatic and renal function
- Nivolumab and ipilimumab are not recommended during pregnancy and in women of childbearing potential not using effective contraception, unless prescribing consultant deems clinical benefit outweighs the potential risk. Effective contraception should be used for at least 5 months following the last dose of nivolumab
- **RCC**
  - ECOG 0-2
  - Histological confirmation of RCC with a clear-cell component
  - Intermediate and poor risk categories as determined by International Metastatic RCC database Consortium (IMDC) study
- **Melanoma**
  - ECOG 0-1
- **MSI-H mCRC**
  - ECOG 0-2
  - Histologically confirmed metastatic or recurrent CRC with dMMR/MSI-H

CAUTIONS:

- Patients with clinically significant autoimmune disease
- Symptomatic CNS metastases
- Immunosuppressive doses of systemic corticosteroids or other immunosuppressive medication(s) (defined as > 10mg prednisoLONE/daily (or steroid equivalent, excluding inhaled or topical steroids)
- Any active clinically significant infection requiring therapy
- **RCC**
  - Ongoing symptomatic cardiac dysrhythmias, uncontrolled atrial fibrillation, or prolongation of the Fridericia corrected QT (QTcF)
  - Poorly controlled hypertension (defined as systolic blood pressure (SBP) of ≥ 150 mmHg or diastolic blood pressure (DBP) of ≥ 90 mmHg), despite antihypertensive therapy

The information contained in this document is a statement of consensus of NCCP and ISMO or IHS professionals regarding their views of currently accepted approaches to treatment. Any clinician seeking to apply or consult these documents is expected to use independent medical judgement in the context of individual clinical circumstances to determine any patient's care or treatment. Use of these documents is the responsibly of the prescribing clinician. and is subject to HSE's terms of use available at http://www.hse.ie/eng/Disclaimer

This information is valid only on the day of printing, for any updates please check www.hse.ie/NCCPSACTRegimens

EXCLUSIONS:

- Hypersensitivity to nivolumab, ipilimumab or to any of the excipients
- Information regarding prior therapy with an anti PD-1 or anti PD-L1 antibody Available on the NCCP website
- Symptomatic interstitial lung disease
- Pregnancy
- Breastfeeding
- **RCC**
  - Prior systemic treatment for advanced renal cell carcinoma

PRESCRIPTIVE AUTHORITY:

The treatment plan must be initiated by a Consultant Medical Oncologist.

TESTS:

Baseline tests:

- FBC, renal and liver profile
- Blood glucose
- Thyroid Function Tests (TFTs)
- Melanoma: BRAF status

Regular tests:

- FBC, renal and liver profile prior to each cycle
- Blood glucose prior to each cycle
- TFTs prior to each cycle
- Cortisol as clinically indicated

Disease monitoring:

Disease monitoring should be in line with the patient's treatment plan and any other test(s) as directed by the supervising Consultant.

DOSE MODIFICATIONS:

- Any dose modification should be discussed with a Consultant.
- Dose escalation or reduction is not recommended. Dosing delay or discontinuation may be required based on individual safety and tolerability
- Management of immune-related adverse reactions may require withholding of a dose or permanent discontinuation of nivolumab in combination with ipilimumab therapy and institution of systemic high-dose corticosteroid

The information contained in this document is a statement of consensus of NCCP and ISMO or IHS professionals regarding their views of currently accepted approaches to treatment. Any clinician seeking to apply or consult these documents is expected to use independent medical judgement in the context of individual clinical circumstances to determine any patient's care or treatment. Use of these documents is the responsibly of the prescribing clinician. and is subject to HSE's terms of use available at http://www.hse.ie/eng/Disclaimer

This information is valid only on the day of printing, for any updates please check www.hse.ie/NCCPSACTRegimens

- If immunosuppression with corticosteroids is used to treat an adverse reaction, a taper of at least 1 month duration should be initiated upon improvement. Rapid tapering may lead to worsening or recurrence of the adverse reaction. Non-corticosteroid immunosuppressive therapy should be added if there is worsening or no improvement despite corticosteroid use. Nivolumab in combination with ipilimumab should not be resumed while the patient is receiving immunosuppressive doses of corticosteroids or other immunosuppressive therapy. Prophylactic antibiotics should be used to prevent opportunistic infections in patients receiving immunosuppressive therapy
- Nivolumab in combination with ipilimumab must be permanently discontinued for:
  - Any severe immune-related adverse reaction that recurs
  - Any life-threatening immune-related adverse reaction
  - Any grade 4 or recurrent grade 3 adverse reactions, persistent grade 2 or 3 adverse reactions despite management
- When nivolumab is administered in combination with ipilimumab, if either agent is withheld, the other agent should also be withheld. If dosing is resumed after a delay, either the combination treatment or nivolumab monotherapy could be resumed based on the evaluation of the individual patient
- Guidelines for withholding of doses or permanent discontinuation are described in Table 3 below

Table 3: Recommended treatment modifications for immune-related adverse events for nivolumab and ipilimumab combination therapy or for administration of the second phase of treatment (nivolumab monotherapy) following combination treatment

Immune-related adverse reaction	Treatment Modification	Corticosteroid Recommendations
Immune-related pneumonitis Patients should be monitored for signs and symptoms of pneumonitis such as radiographic changes (e.g., focal ground glass opacities, patchy filtrates), dyspnoea, and hypoxia. Infectious and disease-related aetiologies should be ruled out		
Grade 2 pneumonitis	Withhold dose(s) until symptoms resolve, radiographic abnormalities improve, and management with corticosteroids is complete	Initiate 1 mg/kg/day methylPREDNISolone equivalents
	If worsening or no improvement occurs despite initiation of corticosteroids, permanently discontinue treatment	Corticosteroid dose should be increased to 2 to 4mg/kg/day methylPREDNISolone equivalents
Grade 3 or 4 pneumonitis	Permanently discontinue treatment	Initiate 2 to 4mg/kg/day methylPREDNISolone equivalents

The information contained in this document is a statement of consensus of NCCP and ISMO or IHS professionals regarding their views of currently accepted approaches to treatment. Any clinician seeking to apply or consult these documents is expected to use independent medical judgement in the context of individual clinical circumstances to determine any patient's care or treatment. Use of these documents is the responsibly of the prescribing clinician. and is subject to HSE's terms of use available at http://www.hse.ie/eng/Disclaimer

This information is valid only on the day of printing, for any updates please check www.hse.ie/NCCPSACTRegimens

Immune-related colitis Patients should be monitored for diarrhoea and additional symptoms of colitis, such as abdominal pain and mucus or blood in stool. Infectious and disease-related aetiologies should be ruled out.		
Grade 2 diarrhoea or colitis	Withhold dose(s) until symptoms resolve and management with corticosteroids, if needed, is complete	Persistent diarrhoea or colitis should be managed with 0.5 to 1mg/kg/day methylPREDNISolone equivalents
	If worsening or no improvement occurs despite initiation of corticosteroids, permanently discontinue treatment	Corticosteroid dose should be increased to 1 to 2mg/kg/day methylPREDNISolone equivalents
Grade 3 or 4 diarrhoea or colitis[a]	Permanently discontinue treatment	Initiate 1 mg/kg/day methylPREDNISolone equivalents
Immune-related hepatitis without HCC Patients should be monitored for signs and symptoms of hepatitis such as transaminase and total bilirubin elevations. Infectious and disease-related aetiologies should be ruled out.		
Grade 2 elevation in aspartate aminotransferase (AST), alanine aminotransferase (ALT), or total bilirubin	Withhold dose(s) until laboratory values return to baseline and management with corticosteroids, if needed, is complete	Persistent elevations in these laboratory values should be managed with 0.5 to 1mg/kg/day methylPREDNISolone equivalents
	If worsening or no improvement occurs despite initiation of corticosteroids, permanently discontinue treatment	Corticosteroid dose should be increased to 1 to 2mg/kg/day methylPREDNISolone equivalents
Grade 3 or 4 elevation in AST, ALT, or total bilirubin	Permanently discontinue treatment	Initiate 1 to 2mg/kg/day methylPREDNISolone equivalents
Immune-related hepatitis with HCC Patients should be monitored for signs and symptoms of hepatitis such as transaminase and total bilirubin elevations. Infectious and disease-related aetiologies should be ruled out.		
If AST/ALT is within normal limits at baseline and increases to > 3 and ≤ 10 times ULN Or Baseline AST/ALT is > 1 and ≤ 3 times ULN and increases to > 5 and ≤ 10 times ULN Or Baseline AST/ALT is > 3 and ≤ 5 times ULN and increases to > 8 and ≤ 10 times ULN	Withhold dose(s) until laboratory values return to baseline and management with corticosteroids, if needed, is complete	Persistent elevations in these laboratory values should be managed with 0.5 to 1mg/kg/day methylPREDNISolone equivalents
	If worsening or no improvement occurs despite initiation of corticosteroids, permanently discontinue treatment	Corticosteroid dose should be increased to 1 to 2mg/kg/day methylPREDNISolone equivalents
AST/ALT increases to > 10 times ULN Or Total bilirubin increases to > 3 times ULN	Permanently discontinue treatment	Initiate 1 to 2mg/kg/day methylPREDNISolone equivalents

The information contained in this document is a statement of consensus of NCCP and ISMO or IHS professionals regarding their views of currently accepted approaches to treatment. Any clinician seeking to apply or consult these documents is expected to use independent medical judgement in the context of individual clinical circumstances to determine any patient's care or treatment. Use of these documents is the responsibly of the prescribing clinician. and is subject to HSE's terms of use available at http://www.hse.ie/eng/Disclaimer

This information is valid only on the day of printing, for any updates please check www.hse.ie/NCCPSACTRegimens

Immune-related nephritis and renal dysfunction Patients should be monitored for signs and symptoms of nephritis or renal dysfunction. Most patients present with asymptomatic increases in serum creatinine. Disease-related aetiologies should be ruled out.		
Grade 2 or 3 creatinine elevation	Withhold dose(s) until creatinine returns to baseline and management with corticosteroids is complete If worsening or no improvement occurs despite initiation of corticosteroids, permanently discontinue treatment	Initiate 0.5 to 1mg/kg/day methylPREDNISolone equivalents Corticosteroid dose should be increased to 1 to 2mg/kg/day methylPREDNISolone equivalents
Grade 4 creatinine elevation	Permanently discontinue treatment	Initiate 1 to 2mg/kg/day methylPREDNISolone equivalents
Immune-related endocrinopathies Patients should be monitored for clinical signs and symptoms of endocrinopathies and for hyperglycaemia and changes in thyroid function (at the start of treatment, periodically during treatment, and as indicated based on clinical evaluation). Patients may present with fatigue, headache, mental status changes, abdominal pain, unusual bowel habits, and hypotension, or nonspecific symptoms which may resemble other causes such as brain metastasis or underlying disease. Unless an alternate aetiology has been identified, signs or symptoms of endocrinopathies should be considered immune-related		
Symptomatic Grade 2 or 3 hypothyroidism, hyperthyroidism, hypophysitis, Grade 2 adrenal insufficiency	Withhold dose(s) until symptoms resolve and management with corticosteroids (if needed for symptoms of acute inflammation) is complete. Hormone replacement or antithyroid medication should be initiated as needed. Treatment should be continued in the presence of hormone replacement therapy/ antithyroid medication as long as no symptoms are present	Initiate 1 to 2mg/kg/day methylPREDNISolone equivalents if acute inflammation is suspected
Grade 4 hypothyroidism Grade 4 hyperthyroidism Grade 4 hypophysitis Grade 3 or 4 adrenal insufficiency	Permanently discontinue treatment	Initiate 1 to 2mg/kg/day methylPREDNISolone equivalents if acute inflammation is suspected
Grade 3 diabetes	Treatment should be withheld, and insulin replacement should be initiated as needed. Monitoring of blood sugar should continue to ensure appropriate insulin replacement is utilised.	N/A
Grade 4 diabetes	Treatment must be permanently discontinued	N/A

The information contained in this document is a statement of consensus of NCCP and ISMO or IHS professionals regarding their views of currently accepted approaches to treatment. Any clinician seeking to apply or consult these documents is expected to use independent medical judgement in the context of individual clinical circumstances to determine any patient's care or treatment. Use of these documents is the responsibly of the prescribing clinician. and is subject to HSE's terms of use available at http://www.hse.ie/eng/Disclaimer

This information is valid only on the day of printing, for any updates please check www.hse.ie/NCCPSACTRegimens

Immune-related skin adverse reactions Rare cases of Stevens-Johnson Syndrome (SJS) and toxic epidermal necrolysis (TEN), some of them with fatal outcome have been observed. If symptoms or signs of SJS or TEN appear, treatment should be discontinued and the patient referred to a specialised unit for assessment and treatment. Caution should be used when considering the use of treatment in a patient who has previously experienced a severe or life-threatening skin adverse reaction on prior treatment with other immune-stimulatory anticancer agents		
Grade 3 rash Grade 4 rash SJS or TEN	Withhold dose(s) until symptoms resolve and management with corticosteroids is complete Permanently discontinue treatment Permanently discontinue treatment	Initiate 1 to 2mg/kg/day methylPREDNISolone equivalents
Immune-related myocarditis Cases of myotoxicity (myositis, myocarditis, and rhabdomyolysis), some with fatal outcome, have been reported with nivolumab or nivolumab in combination with ipilimumab. If a patient develops signs and symptoms of myotoxicity, close monitoring should be implemented, and the patient referred to a specialist for assessment and treatment without delay. The diagnosis of myocarditis requires a high index of suspicion. Patients with cardiac or cardio-pulmonary symptoms should be assessed for potential myocarditis.		
Grade 2 myocarditis Grade 3 or 4 myocarditis	Withhold dose(s) until symptoms resolve and management with corticosteroids is complete[b] Permanently discontinue treatment	If myocarditis is suspected, prompt initiation of methylPREDNISolone 1 to 2mg/kg/day equivalents and prompt cardiology consultation with diagnostic workup according to current clinical guidelines should be initiated
Other immune-related adverse reactions The following immune-related adverse reactions were reported in less than 1% of patients treated with nivolumab monotherapy or nivolumab in combination with ipilimumab in clinical trials across doses and tumour types: pancreatitis, uveitis, demyelination, autoimmune neuropathy (including facial and abducens nerve paresis), Guillain-Barré syndrome, myasthenia gravis, Myocarditis-Myositis Myasthenia Gravis Overlap Syndrome, myasthenic syndrome, aseptic meningitis, encephalitis, gastritis, sarcoidosis, duodenitis, myositis, myocarditis, rhabdomyolysis, and myelitis. Cases of Vogt-Koyanagi-Harada syndrome, hypoparathyroidism, and cystitis noninfective have been reported post-marketing. For suspected immune-related adverse reactions, adequate evaluation should be performed to confirm aetiology or exclude other causes		
Grade 3 (first occurrence) Grade 4 or recurrent Grade 3; persistent Grade 2 or 3 despite treatment modification; inability to reduce corticosteroid dose to 10mg prednisoLONE or equivalent per day	Withhold dose(s) Permanently discontinue treatment	Corticosteroids initiated as clinically indicated
Myocarditis-Myositis-Myasthenia Gravis Overlap Syndrome[c] Cases of Myocarditis-Myositis-Myasthenia Gravis Overlap Syndrome (presenting as an overlap of either two or all three conditions), some with fatal outcome, have been reported with nivolumab and nivolumab in combination with other therapeutic agents. Early recognition and aggressive management are essential to address associated morbidity and risk of mortality.		

The information contained in this document is a statement of consensus of NCCP and ISMO or IHS professionals regarding their views of currently accepted approaches to treatment. Any clinician seeking to apply or consult these documents is expected to use independent medical judgement in the context of individual clinical circumstances to determine any patient's care or treatment. Use of these documents is the responsibly of the prescribing clinician. and is subject to HSE's terms of use available at http://www.hse.ie/eng/Disclaimer

This information is valid only on the day of printing, for any updates please check www.hse.ie/NCCPSACTRegimens

Grade 2 Myocarditis-Myositis-Myasthenia Gravis Overlap Syndrome	Withhold dose(s) until symptoms resolve and management with corticosteroids is complete If worsening or no improvement occurs despite initiation of corticosteroids, treatment must be permanently discontinued	Corticosteroids initiated as clinically indicated Corticosteroid dose should be adjusted as clinically indicated
Grade 3 or 4 Myocarditis-Myositis-Myasthenia Gravis Overlap Syndrome	Permanently discontinue treatment	Corticosteroids initiated as clinically indicated
Infusion reactions		
Mild or moderate	May receive treatment with close monitoring and use of premedication according to local treatment guidelines for prophylaxis of infusion reactions	
Severe or life-threatening infusion reaction	Discontinue infusion	Administer appropriate medical therapy
Note: Toxicity grades are in accordance with National Cancer Institute Common Terminology Criteria for Adverse Events Version 4.0 (NCI-CTCAE v4). [a]During administration of the second phase of treatment (nivolumab monotherapy) following combination treatment, permanently discontinue treatment if Grade 3 diarrhoea or colitis occurs. [b]The safety of re-initiating nivolumab or nivolumab in combination with ipilimumab therapy in patients previously experiencing immune-related myocarditis is not known. [c]Presenting as an overlap of either two or all three conditions. The most severe CTCAE grade from the individual events should be considered to assess the recommended treatment modification. **Table 4** has been adapted from nivolumab and ipilimumab SmPCs.		

Renal and Hepatic Impairment:

Table 4: Dose modification of nivolumab and ipilimumab in renal and hepatic impairment

Drug	Renal Impairment		Hepatic Impairment	
Nivolumab	**Mild-Moderate**	No dose adjustment necessary	**Mild**	No dose adjustment necessary
	Severe	Has not been studied	**Moderate -Severe**	Has not been studied. Nivolumab must be administered with caution in patients with: • moderate (total bilirubin >1.5x to 3x ULN and any AST) or • severe (total bilirubin >3x ULN and any AST)
Ipilimumab	No specific dose adjustment is necessary in patients with mild to moderate renal dysfunction.		No specific dose adjustment is necessary in patients with mild hepatic impairment. Administer with caution in patients with transaminase levels ≥ 5 x ULN or bilirubin levels > 3 x ULN at baseline.	

The information contained in this document is a statement of consensus of NCCP and ISMO or IHS professionals regarding their views of currently accepted approaches to treatment. Any clinician seeking to apply or consult these documents is expected to use independent medical judgement in the context of individual clinical circumstances to determine any patient's care or treatment. Use of these documents is the responsibly of the prescribing clinician. and is subject to HSE's terms of use available at http://www.hse.ie/eng/Disclaimer

This information is valid only on the day of printing, for any updates please check www.hse.ie/NCCPSACTRegimens

SUPPORTIVE CARE:

EMETOGENIC POTENTIAL:

- As outlined in NCCP Classification Document for Systemic Anti-Cancer Therapy (SACT) Induced Nausea and Vomiting - Available on the NCCP website

Nivolumab: Minimal **(Refer to local Policy)**
Ipilimumab: Low **(Refer to local policy)**
For information:
Within NCIS regimens, antiemetics have been standardised by Medical Oncologists and Haemato-oncologists and information is available in the following documents:

- NCCP Supportive Care Antiemetic Medicines for **Inclusion in NCIS** (Medical Oncology) - Available on the NCCP website
- NCCP Supportive Care Antiemetic Medicines for **Inclusion in NCIS** (Haemato-oncology) - Available on the NCCP website

PREMEDICATIONS: Not usually required

OTHER SUPPORTIVE CARE: No specific recommendations

ADVERSE EFFECTS / REGIMEN SPECIFIC COMPLICATIONS:

The adverse effects listed are not exhaustive. Please refer to the relevant Summary of Product Characteristics for full details.

- **Cardiac adverse events and pulmonary embolism**: Patients should be monitored for cardiac and pulmonary adverse reactions continuously, as well as for clinical signs, symptoms, and laboratory abnormalities indicative of electrolyte disturbances and dehydration prior to and periodically during combination treatment. Nivolumab in combination with ipilimumab should be discontinued for life-threatening or recurrent severe cardiac and pulmonary adverse reactions
- **Immune related adverse reactions:** Please see Table 3 for dose modifications of nivolumab with ipilimumab combination or for nivolumab monotherapy following combination treatment

DRUG INTERACTIONS:

- Current drug interaction databases should be consulted for more information.

COMPANY SUPPORT RESOURCES/USEFUL LINKS:

Please note that this is for information only and does not constitute endorsement by the NCCP

Patient Alert Card:
Nivolumab: https://assets.hpra.ie/products/Human/10066/c02753be-51a5-44fd-8117-123823bdcff8.pdf

The information contained in this document is a statement of consensus of NCCP and ISMO or IHS professionals regarding their views of currently accepted approaches to treatment. Any clinician seeking to apply or consult these documents is expected to use independent medical judgement in the context of individual clinical circumstances to determine any patient's care or treatment. Use of these documents is the responsibly of the prescribing clinician. and is subject to HSE's terms of use available at http://www.hse.ie/eng/Disclaimer

This information is valid only on the day of printing, for any updates please check www.hse.ie/NCCPSACTRegimens

Ipilimumab: https://assets.hpra.ie/products/Human/9440/0781c3d7-ff8d-4cc7-9f0a-80cf9a10e59f.pdf

Patient Information Guide:
Ipilimumab: https://assets.hpra.ie/products/Human/9440/2f064c72-ccef-492b-a068-bc72d8b522cf.pdf

REFERENCES:

1. Motzer RJ et al. Nivolumab plus Ipilimumab versus Sunitinib in Advanced Renal-Cell Carcinoma. NEJM. 2018 Apr 5; 378(14):1277-1290. doi: 10.1056/NEJMoa1712126. Epub 2018 Mar 21. Including supplementary material.
2. NCCP SACT Melanoma Clinical Advisory Group: Alternate Dosing Posology of Nivolumab 3mg/kg Ipilimumab 1mg/kg in melanoma: Evidence into practice – rapid review September 2020.
3. Lenz HJ, et al. First-Line Nivolumab Plus Low-Dose Ipilimumab for Microsatellite Instability-High/Mismatch Repair- Deficient Colorectal Cancer: The Phase II CheckMate 142 Study. J Clin Oncol 2021; 40(2):161-170.
4. Aron M, Nguyen MM, Stein RJ, Gill IS. Impact of Gender in Renal Cell Carcinoma: An Analysis of the SEER Database. European Urology. 2008;54(1):133-42.
5. NCCP Classification Document for Systemic Anti-Cancer Therapy (SACT) Induced Nausea and Vomiting. V6 2025. Available at: https://www.hse.ie/eng/services/list/5/cancer/profinfo/chemoprotocols/nccp-classification-document-for-systemic-anti-cancer-therapy-sact-induced-nausea-and-vomiting.pdf
6. Ipilimumab (YERVOY®) Summary of Product Characteristics. Last updated 23/07/2021. Accessed April 2023. Available at: https://www.ema.europa.eu/en/documents/product-information/yervoy-epar-product-information_en.pdf
7. Nivolumab (OPDIVO®) 10mg/mL Concentrate for Solution for Infusion. Nivolumab (OPDIVO®) 600mg Solution for Injection. Summary of Product Characteristics (EMA). Last updated: 23/06/2026. Accessed 24/06/2026. Available at: https://www.ema.europa.eu/en/documents/product-information/opdivo-epar-product-information_en.pdf

Version	Date	Amendment	Approved By
1	21/08/2019		Prof Maccon Keane
2	09/10/2019	Updated adverse effects/regimen specific complications section as per SmPC update regarding CMV infection/reactivation	Prof Maccon Keane
3	23/09/2020	Addition of melanoma indication	Prof Maccon Keane

The information contained in this document is a statement of consensus of NCCP and ISMO or IHS professionals regarding their views of currently accepted approaches to treatment. Any clinician seeking to apply or consult these documents is expected to use independent medical judgement in the context of individual clinical circumstances to determine any patient's care or treatment. Use of these documents is the responsibly of the prescribing clinician. and is subject to HSE's terms of use available at http://www.hse.ie/eng/Disclaimer

This information is valid only on the day of printing, for any updates please check www.hse.ie/NCCPSACTRegimens

4	01/02/2021	Update reimbursement status	Prof Maccon Keane
5	22/10/2021	Added to baseline tests. Updated dose modifications for adverse events and hepatic impairment. Updated emetogenic potential. Updated adverse effects/regimen specific complications section.	Prof Maccon Keane
6	01/06/2023	New indication added	Prof Maccon Keane
6a	08/05/2025	Update to ICD-10 code	NCCP
6b	14/08/2025	Updated exclusions section with information regarding previous PDL1 treatment. Added links to emetogenic potential section.	NCCP
7	11/02/2026	Updated to include subcutaneous administration and dosing for nivolumab (Table 2). Updated cautions section. General template and formatting updates.	Prof Maccon Keane
7a	20/08/2026	Amendment made to treatment section relating to dMMR/MSI-H mCRC indication 00551c: timeframe between finishing combination therapy and initiation of monotherapy (correction to 3-weeks only). Combined Table 3 and Table 5 (adverse events tables)	Prof. Maccon Keane

Comments and feedback welcome at oncologydrugs@cancercontrol.ie.

[i] The administration of nivolumab 3mg/kg in combination with ipilimumab 1mg/kg is an unlicensed dosing posology for this indication in Ireland. Patients should be informed of this and consented to treatment in line with the hospital's policy on the use of unlicensed medication and unlicensed or "off label" indications. Prescribers should be fully aware of their responsibility in communicating any relevant information to the patient and also ensuring that the unlicensed or "off label" means of administration has been acknowledged by the hospital's Drugs and Therapeutics Committee, or equivalent, in line with hospital policy.

The information contained in this document is a statement of consensus of NCCP and ISMO or IHS professionals regarding their views of currently accepted approaches to treatment. Any clinician seeking to apply or consult these documents is expected to use independent medical judgement in the context of individual clinical circumstances to determine any patient's care or treatment. Use of these documents is the responsibly of the prescribing clinician. and is subject to HSE's terms of use available at http://www.hse.ie/eng/Disclaimer

This information is valid only on the day of printing, for any updates please check www.hse.ie/NCCPSACTRegimens